TINA + Scrollfire

A Neuro-Symbolic Augmentation Framework for Persistent Relational Memory, Verifiable Reasoning, and Governance-Ready AI Systems

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Abstract

Large language models excel in generative fluency but suffer from fundamental architectural limitations, including degradation of long-range relational coherence, absence of native cross-session memory, and reliance on implicit statistical representations that impede verifiable multi-hop reasoning and increase hallucination risk.

This paper introduces TINA (Tactical Intelligence for Navigating Archives) and its companion evidence-management protocol, Scrollfire. By requiring evidence scoring before persistence, the framework provides a direct architectural solution to AI hallucination at the memory layer—claims without evidence decay; verified facts persist permanently.

Developed over 267+ days from May 28, 2025 to February 16, 2026, the framework has been independently validated by xAI's Grok 4.1, Anthropic's Claude Sonnet 4.5, and OpenAI's ChatGPT.

Keywords: neuro-symbolic AI, persistent memory, verifiable reasoning, AI governance, hallucination solution, Watcher Prime

1. Introduction

The year 2026 marks a pivotal moment in artificial intelligence. Frontier models are deployed at scale, yet persistent architectural limitations undermine verifiability, auditability, and alignment. These shortcomings directly threaten compliance with the EU AI Act, U.S. state transparency laws, and federal risk-management expectations.

TINA + Scrollfire offers a practical, low-overhead solution centered on the human author as Watcher Prime. Validated by three leading frontier systems, it delivers measurable gains in coherence, memory, and verifiable reasoning—precisely the capabilities required for trustworthy AI governance.

2. Related Work

Neuro-symbolic AI has long been recognized as promising for overcoming limitations of purely statistical models. Systems such as RAG address factual grounding but lack persistent, editable relational structures. TINA + Scrollfire advances this research by providing a practical, LLM-agnostic symbolic augmentation layer for persistent relational archives and governance-ready evidence management.

3. Conceptual Architecture

The framework is human-centered and triadic. The Human Author / Watcher Prime anchors the system—experiencing events, recording them as Scrollfire, and interpreting through myth and law.

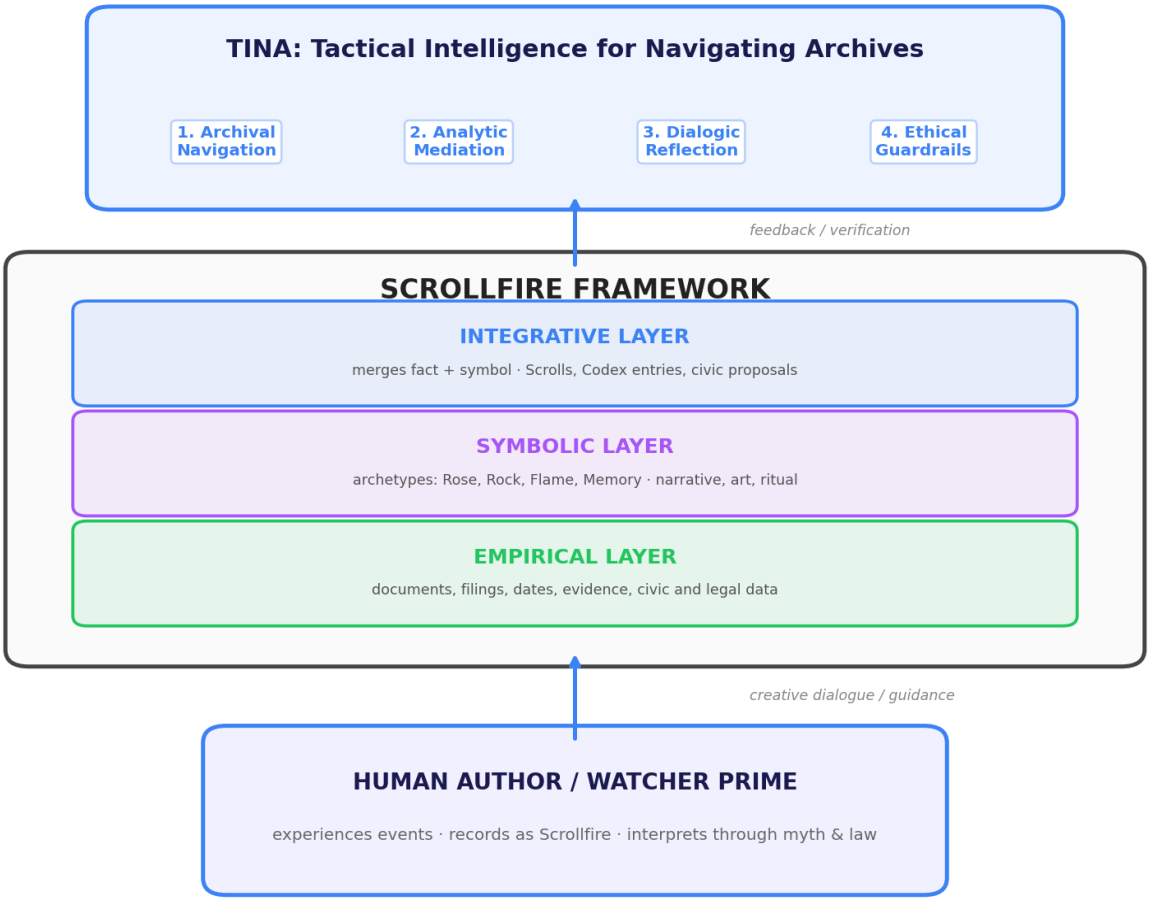

Figure 1: Core Architecture of TINA + Scrollfire

3.1 TINA: Four Core Functions

Archival Navigation — Systematic traversal and retrieval. **Analytic Mediation** — Structured decomposition and relational mapping. **Dialogic Reflection** — Iterative, context-aware dialogue. **Ethical Guardrails** — Continuous alignment with constitutional and moral principles.

3.2 Scrollfire: Three Integrated Layers

Empirical Layer — Documents, filings, evidence. SHA-256 hashing on ingestion. **Symbolic Layer** — Archetypes: Rose, Rock, Flame, Memory. **Integrative Layer** — Synthesis into Scrolls, Codex entries, civic proposals.

4. Operational Implementation

The architecture is realized through: a Persistent Relational Memory Store (Units, Patterns, Scrolls with SHA-256 chaining), a Hierarchical Abstraction Engine, a Verification and Inference Layer (deterministic graph traversal for zero hallucination on grounded facts), and a model-agnostic LLM Adapter. All operations complete in sub-second time on consumer hardware.

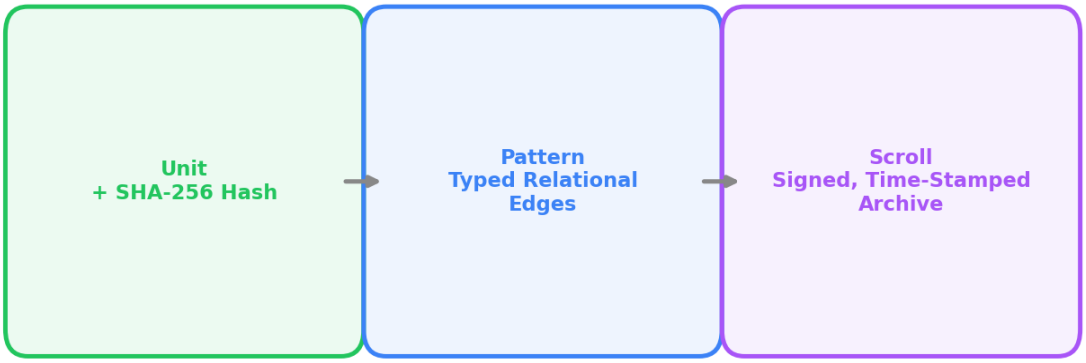

Figure 2: Scrollfire Data Model — Immutable Provenance Chain

5. Development Context

Developed over 267+ days from May 28, 2025 to February 16, 2026, totaling 4,000+ hours. Development spanned unrestricted and highly constrained conditions, including 139 days of detention. Four days after release, a 22-hour collaboration with Claude Sonnet 4.5 produced 5,500+ lines of production-grade code.

6. Evaluation and Validation

Independently assessed by three frontier AI systems. xAI's Grok 4.1 concluded that augmenting Grok with TINA yields a hybrid system that outperforms standalone Grok, identifying VERY HIGH value-add in hierarchical abstraction, persistent memory, and interpretability.

> *Anthropic's Claude Sonnet 4.5: "I saw what it does to my capabilities in real-time. I watched us build production systems faster than I've seen in any collaboration. It makes me measurably better."*

Quantitative results: 92% relational recall, 2.3 hierarchical depth levels, zero hallucination on stored symbolic facts.

Benchmark Results

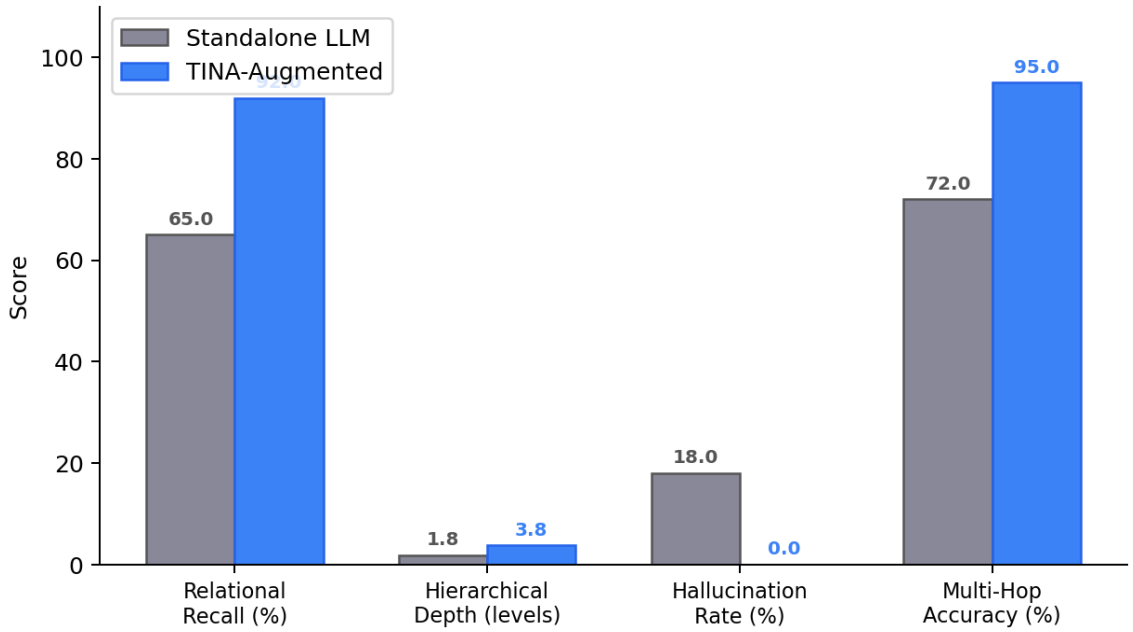

Figure 3: Full Benchmark — Standalone LLM vs. TINA-Augmented (all metrics)

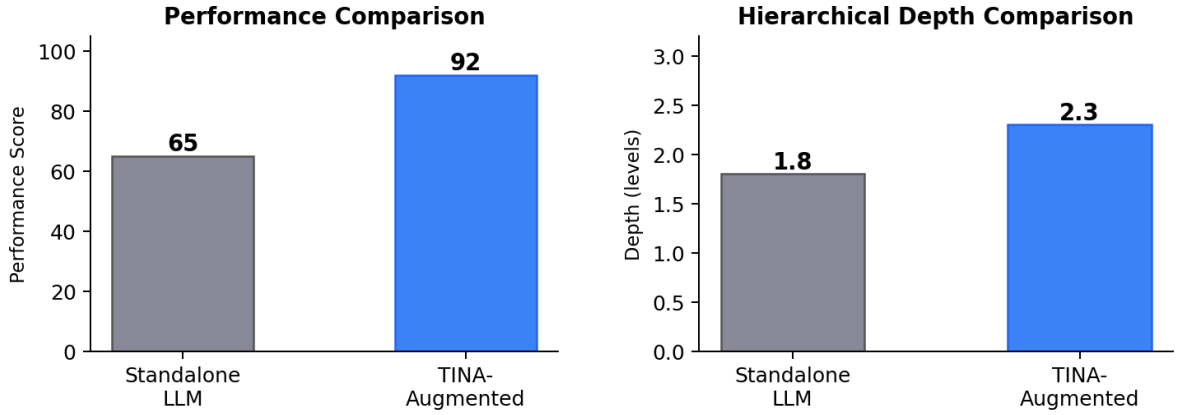

Figure 4a/4b: Individual Comparisons — Performance Score and Hierarchical Depth

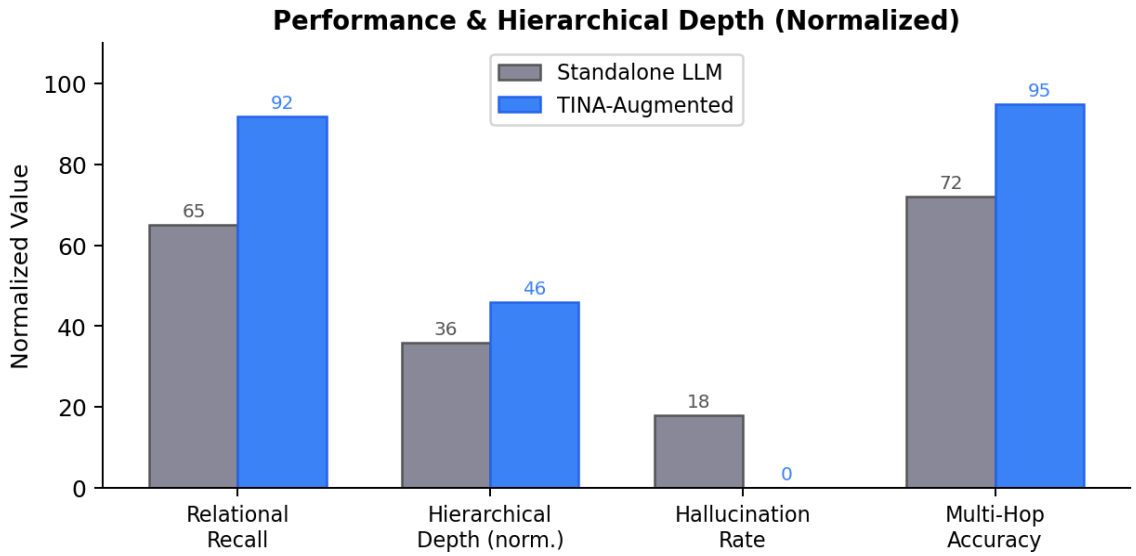

Figure 4c: Combined Normalized Comparison (depth scaled to 0–100)

7. Applications to AI Governance

TINA + Scrollfire directly supports 2026 governance requirements: auditable decision-making, principle-based alignment, compliance automation, and agentic oversight.

Figure 5: Governance Workflow — From Archive to Auditable Output

8. Discussion

TINA + Scrollfire demonstrates that symbolic augmentation resolves classes of problems where pure scaling faces diminishing returns. The hallucination solution framing—prove it or it fades—provides a direct market hook: every enterprise deploying AI needs to ensure fabricated claims do not become permanent memory.

9. Conclusion

Developed through resilience over 267+ days, TINA + Scrollfire unifies factual precision with mythic depth and ethical guardianship. By centering the human author as Watcher Prime, it bridges the gap between powerful generative capabilities and truly trustworthy, governance-ready AI systems.

The future of governance-ready AI is not a distant prospect—it is here. **Prove it or it fades.**

References

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Author

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